# Reproducible Literature Synthesis with infrastructure/search and infrastructure/reference

A configurable pipeline from query → references.bib → LLM-driven reading report

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DOI: 10.5281/zenodo.21298894

November 9, 2026

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# 1 Abstract

This paper documents `template_search_project`, the literature-search exemplar shipped with the Research Project Template. The project demonstrates **two configurable, reproducible pipelines** sharing the same configuration file and the same `infrastructure/search/` + `infrastructure/reference/` modules. The standard pipeline (`scripts/run_search_pipeline.py`) handles a single `SearchQuery` end-to-end. The deep-search pipeline (`scripts/run_deep_search.py`, see sec. 8) fans out across a list of keywords (each capped at 100 papers per keyword from `project_config.deep_search.max_results_per_keyword` in `manuscript/config.yaml`), fully enriches every paper with its abstract and PDF fulltext, and (optionally) uses the local LLM to write a multi-section reading note for every paper. When a deep-search aggregate exists, the latest run covered **3** keyword(s) with unique paper(s) after cross-keyword deduplication. Both turn a free-text topic into:

1. a deduplicated, year-filtered set of papers drawn from arXiv, Crossref, optional local corpora, and (opt-in) Paperclip;
2. a Pandoc-compatible `references.bib` byte-identical in style to the canonical exemplar in `template_code_project` (file `manuscript/references.bib`);
3. cached abstracts and (optionally) extracted PDF full text, written to disk under stable per-paper identifiers; and
4. an LLM-synthesised reading report assembled from per-paper analyses and a cross-corpus thematic synthesis, all produced by a local Ollama model with pinned seed and temperature.

All discovery logic lives in `infrastructure/search/literature/` (source on GitHub); all export logic lives in `infrastructure/reference/citation/` (source on GitHub); LLM synthesis reuses the existing `infrastructure/llm/` (source on GitHub) bridge. The project itself contains only thin orchestration, manuscript prose, and a test suite — perfectly mirroring the **two-layer architecture** the template enforces.

The motivating concern is *reproducibility*: a query at time $t_0$ should produce the same results at time $t_1$ unless the cache is explicitly invalidated. This is achieved by deterministic search caching keyed on canonical query identity, on-disk caching of every fetched abstract / PDF, and pinned LLM seeds. The same `manuscript/config.yaml` that drives the pipeline is also the only configuration any reviewer needs.

**Run snapshot.** With the bundled `manuscript/config.yaml`, the most recent pipeline execution evaluated the query *"reproducible research optimization"* against local, returned 6 deduplicated paper(s) (4 carrying a DOI, 6 carrying an abstract), and recorded backend errors: none. Resolve `{{…}}` tokens by running `scripts/z_generate_manuscript_variables.py` after `run_search_pipeline.py`; the script writes `output/data/manuscript_variables.json` and resolved markdown under `output/manuscript/`, which the PDF-rendering stage prefers when present.

**Claim boundary.** The committed `data/corpus.json` and the default local-source run are deterministic workflow fixtures. Their counts, summaries, and generated bibliography exercise the pipeline contract; they are not empirical findings about the literature.

**Keywords:** literature search, BibTeX automation, reproducible research, local LLM synthesis, scientific infrastructure

## 2 Introduction

Reproducible computational research demands that every claim be traceable back to a stable artifact — code, data, and citations alike [Peng, 2011]. Manual literature curation is a well-known bottleneck in such workflows: a graduate student writing a related-work section may spend hours searching arXiv, Crossref, and Google Scholar; copying citations into a `.bib` file by hand; and tracking which papers they have actually read. Three failure modes recur:

1. **Style drift** — hand-edited `.bib` files accumulate formatting inconsistencies that hide real semantic conflicts in version-control diffs.
2. **Stale state** — the bibliography, the reading list, and the manuscript prose drift apart as the project evolves; the citation key in the manuscript no longer matches the entry in `.bib`, or the entry no longer matches the actual paper.
3. **Lost context** — abstracts and full text are read once during search, then discarded; six months later the same paper has to be re-skimmed to recall its contribution.

`template_search_project` exists to demonstrate one disciplined solution. The pipeline outputs are summarised in sec. 3 (overview figure at the start of that section):

- The discovery side (`infrastructure/search/`) provides multi-source paper search with failure-isolated aggregation, DOI/arXiv-aware deduplication, and deterministic JSON caching keyed on canonical query identity.
- The export side (`infrastructure/reference/`) provides BibTeX read/write/convert facilities byte-compatible with the existing exemplar `references.bib`, suitable for the combined-PDF pipeline (Pandoc `--natbib` + BibTeX).
- A small project-local synthesis layer (in `src/pipeline/synthesis.py`) takes enriched papers, builds reproducible LLM prompts, and assembles a markdown reading report.

The project is *configurable* via a single `manuscript/config.yaml`: changing the topic, year filters, backend set, enrichment level, and LLM parameters never requires editing code. The project is *modular* in the strict sense the template uses: every reusable component lives in `infrastructure/`, and `src/` contains only project-specific orchestration.

The contribution of this exemplar is therefore not a new algorithm; it is a **demonstration that a reproducible literature workflow can be built from existing template infrastructure** with no new optional dependencies, no mocks in the test suite, and complete configurability through a single YAML file.

# 3 Methodology

Two distinct workflows run on top of `infrastructure/search/literature` and `infrastructure/reference/citation`:

- **Standard pipeline** (`scripts/run_search_pipeline.py` → `src/pipeline/pipeline.py::run_literature_pipeline`) — single `SearchQuery`. Four pure-orchestration stages with no LLM dependency: (1) search via `LiteratureClient`, (2) enrichment via `AbstractFetcher` and (optional) `FulltextFetcher`, (3) collision-free citation-key generation in `_build_citation_keys`, (4) writing `output/corpus.json` + `manuscript/references.bib` + `output/enrichment_log.json`. The orchestrator script then optionally calls `src/pipeline/synthesis.py` for per-paper and corpus LLM synthesis and `src/analysis/report.py` for the final reading report.
- **Deep search** (`scripts/run_deep_search.py` → `src/search/deep_search.py::run_deep_search`) — multi-keyword fan-out: each keyword runs its own `SearchQuery` capped at `max_results_per_keyword` (100 by default), every paper is fully enriched (abstract + PDF fulltext when available), and an LLM-driven multi-section deep summary (CONTRIBUTION / METHOD / EVIDENCE / LIMITATIONS / CONNECTIONS / SIGNIFICANCE / TAGS) is written for each paper as a standalone markdown reading note. Output lands under `output/deep_search/<keyword_slug>/` plus aggregate `aggregate.json`, `aggregate_report.md`, and a unified, deduplicated `manuscript/references_deep.bib` with collision-free citation keys.

The standard pipeline is described first in this section; the deep-search workflow is documented in sec. 8. Diagnostic figures for the latest pipeline run appear at the end of this section.

## 3.1 Search

The search stage is intentionally faithful to the standard literature-search pattern documented in foundational optimisation textbooks [Boyd and Vandenberghe, 2004, Nocedal and Wright, 2006] — a deterministic query, capped result count, and explicit failure isolation between sources — so reviewers familiar with those references can reason about the workflow without learning new abstractions.

A `SearchQuery` is constructed from `config.search`:

```
SearchQuery(
    text=config.search.query,
    max_results=config.search.max_results,
    year_min=config.search.year_min,
    year_max=config.search.year_max,
)
```

A `LiteratureClient` is constructed with the configured backends. Each backend produces a normalised `Paper` record; the aggregator deduplicates by DOI → arXiv id → normalised (title, year), keeping the highest-scored copy and filling missing fields from the loser.

Per-backend errors are recorded into `SearchResult.errors` rather than raised. A network outage in one backend never breaks the workflow; partial coverage is reported by the final stage.

## 3.2 Cache

`SearchCache` writes one JSON file per query, named by a 16-character SHA-256 prefix of the canonical query identity. Identical queries (modulo whitespace and case) share a cache entry. Cache files are pretty-printed JSON, version-control-friendly, and contain a `_cached_at` timestamp for optional TTL enforcement.

## 3.3 Enrichment

Two fetchers populate fields the search backends did not supply:

- `AbstractFetcher` — currently fetches arXiv abstracts via the export API, writes them to `<safe_id>.txt` under the configured cache directory, and re-uses them on subsequent runs.
- `FulltextFetcher` — downloads PDFs (arXiv URL, `paper.pdf_url`, or a caller-supplied override), writes the bytes verbatim to `<safe_id>.pdf`, and extracts text via `pypdf` to `<safe_id>.txt`. Without `pypdf` the PDF is still cached, and the fetcher returns `status="error"` with an informative message; the rest of the pipeline continues.

Both fetchers stamp `paper.abstract` / `paper.fulltext` in place, so downstream stages see enriched records without re-loading.

### 3.4 Export

For every paper, `paper_to_bibentry()` produces a `BibEntry` whose:

- citation key follows the exemplar's `<author><year><title-word>` convention with stop-word filtering and unicode folding;
- entry type is routed by `venue_type` (journal → `@article`, conference → `@inproceedings`, book → `@book`, preprint → `@article`, etc.);
- fields are emitted in the order observed in `references.bib`: title, author, journal/booktitle, year, volume, number, pages, publisher, edition, isbn, doi, url, abstract, keywords.

A `BibDatabase` collects these entries and `write_bibfile` renders them in the project's house format: 2-space indent, trailing-comma rule, `pages={N--M}`, verbatim DOIs/years, bare unicode.

### 3.5 Synthesis

Two LLM passes produce the reading report (see `src/pipeline/synthesis.py`):

- **Per-paper synthesis** — `build_paper_block(paper, citation_key, max_fulltext=4000)` renders the paper as a markdown block; `synthesise_per_paper` formats `PROMPT_PER_PAPER` and calls the injected `llm` callable. The prompt requests five sections: CONTRIBUTION, METHOD, EVIDENCE, LIMITATION, TAGS, plus a citation-key reference.
- **Corpus synthesis** — `build_corpus_block` concatenates every paper into a single citation-keyed block; `synthesise_corpus` formats `PROMPT_CORPUS`, which asks for 3–7 thematic clusters, methodological agreements / disagreements (>= 2 papers each), and three open questions that the corpus does not answer.

Both functions return a `SynthesisResult(kind, prompt, text, paper_id)` record so the prompt is recoverable for reproducibility. The synthesis layer takes a callable `llm: (str) -> str` so tests pass a deterministic local function (no Ollama dependency) and runtime callers pass a thin adapter around `infrastructure.llm.LLMClient`. Determinism in production runs is enforced by `OllamaClientConfig(seed=42, temperature=0.0)`.

The deep-search workflow uses a richer prompt (`src/search/deep_search.py::DEEP_PROMPT`) with seven sections (CONTRIBUTION / METHOD / EVIDENCE / LIMITATIONS / CONNECTIONS / SIGNIFICANCE / TAGS) and a much larger `max_fulltext` budget (400000 characters by default).

### 3.6 Report

`src/analysis/report.py::write_reading_report` assembles a markdown file with:

- Topic, result count, year filter, and any backend errors at the top.
- A per-source count table.
- One-line summaries for every paper.
- The corpus synthesis (if present).
- All per-paper notes (if present).

Citation keys appear in `[brackets]` so a downstream tool — for example a Pandoc filter or a manual search — can resolve them against the auto-generated `references.bib`.

### 3.7 Diagnostic figures

`scripts/y_generate_search_figures.py` (a thin orchestrator over `src/publish/figures.py`) writes three diagnostic plots into `../figures/` from `output/search/results.json`. Each figure uses Matplotlib's `Agg` backend so the pipeline runs headlessly in CI; the colour palette is colourblind-safe (Wong, *Nature Methods* 2011).

fig. 1 reports the per-backend contribution counts before deduplication, surfacing which sources actually returned coverage for the configured query. The bar values are read directly from `SearchResult.per_source_counts` (set by `LiteratureClient` *before* the DOI / arXiv-id / title merge step), so a backend that returned five papers all duplicating arXiv hits still scores five here.

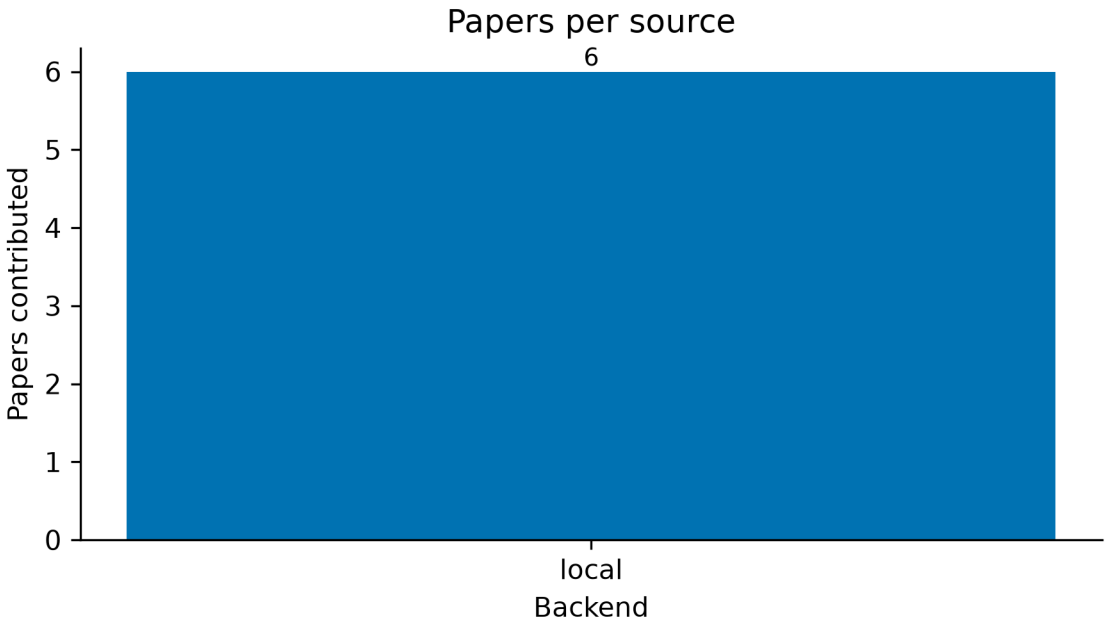



Figure 1: Per-source paper counts read from `SearchResult.per_source_counts` (pre-deduplication contribution per backend). The numeric label above each bar reports the raw count; the y-axis spans `[0, max + headroom]`. Bar order follows `project_config.search.sources`. Empty runs render `(no results)` centred. Generated by `src/publish/figures.py::plot_papers_per_source`.

fig. 2 shows the publication-year distribution *after* the merge step (one bar per unique paper, not per backend hit) — useful for spotting backend coverage gaps in older / newer literature. Papers with no `year` field are dropped silently from the histogram (they remain in the corpus).

fig. 3 shows the per-paper relevance scores returned by the backends, ranked descending. The offline `LocalBackend` sets `Paper.score` to the fraction of query terms found in a paper's searchable title, abstract, keywords, or author text. This is a lexical match score, not a calibrated relevance probability. A backend that leaves `Paper.score` at its default `0.0` produces a zero-length bar while retaining the paper's y-axis tick.

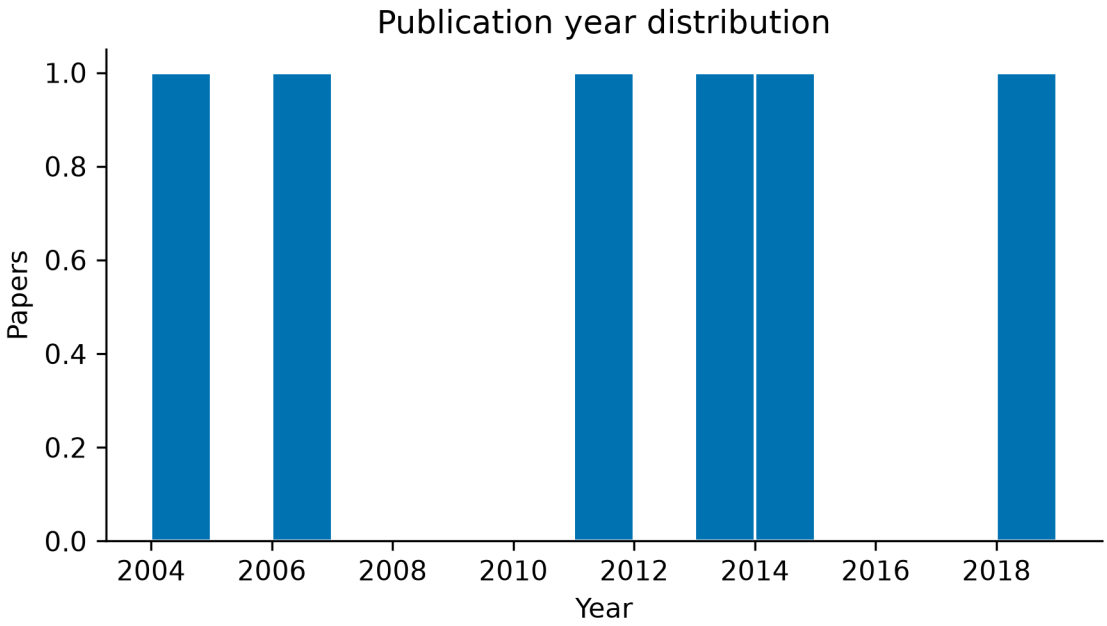



Figure 2: Publication-year histogram of the deduplicated paper roster. One bin per year (no smoothing); the x-axis spans the observed `[min(year), max(year)]` from `result.papers`. Papers with `year is None` are dropped; the y-axis is per-year paper count. Generated by `src/publish/figures.py::plot_year_histogram`.

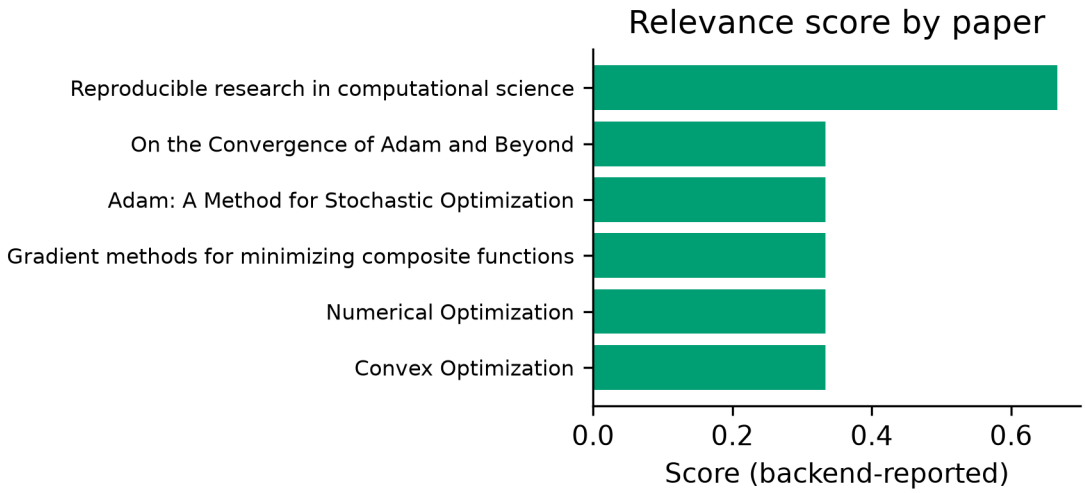



Figure 3: Per-paper backend-reported relevance scores ranked descending (highest at top). Each horizontal bar is one `Paper.score`; long y-tick titles are truncated with an ellipsis. `LocalBackend` scores are query-term match fractions; an unspecified score defaults to `0.0` and yields a zero-length bar. Generated by `src/publish/figures.py::plot_score_distribution`.

# 4 Results

**Run snapshot.** With the bundled `manuscript/config.yaml` the most recent execution evaluated the query *"reproducible research optimization"* against local, returned 6 deduplicated paper(s) (4 carrying a DOI, 6 carrying an abstract); the per-source breakdown is local=6 and recorded backend errors are none. The deep-search workflow (sec. 8) covered 3 keyword(s) — *convex optimization; stochastic gradient descent; reproducible research* — drawn from , producing unique paper(s) after cross-keyword deduplication.

When `sources: [local]` is used, this section reports fixture execution only. It must not be read as a claim about empirical literature coverage; a live-provider run requires source-level provenance and a separately reviewed claim boundary.

The diagnostic figures generated for this run are catalogued in sec. 3: fig. 1 surfaces per-backend coverage, fig. 2 surfaces the temporal distribution, and fig. 3 surfaces the relevance-score profile. The full determinism contract for each stage is itemised in tbl. 1 of sec. 7.

## 4.1 Interpreting the run snapshot

The numerical values in the run-snapshot paragraph that opens this section are read directly from `output/run_summary.json` and `output/data/manuscript_variables.json` so they always reflect the most recent run rather than a stale claim hand-typed into the prose. Three properties are worth highlighting (the formal determinism contract for each underlying stage is enumerated in tbl. 1):

- **Cache reuse is observable.** A second invocation against the same config produces a byte-identical artifact tree (modulo the wall-clock timestamp inside `output/search/cache/search_<hash>.json` itself); see also the *Search (cached hit)* row of tbl. 1 and the verification recipe in sec. 7. The cache file thus doubles as a cryptographic seal: re-running is a file read, not a network round-trip.
- **Deduplication is signal-preserving.** The aggregator merges by DOI → arXiv id → normalised (title, year), keeping the highest-scored copy and filling missing fields from the loser (see *Dedup / merge* in tbl. 1 and the per-backend pre-dedup view in fig. 1). The `RESULT_NUM_PAPERS` figure therefore equals "papers a reviewer needs to read", not "raw backend hit count" — the per-source contributions in `RESULT_PER_SOURCE` are the pre-dedup view.
- **Enrichment coverage is honest.** `RESULT_WITH_ABSTRACT` and `RESULT_WITH_DOI` count fields the corpus or the `AbstractFetcher` actually populated, never values inferred. When a paper is missing a DOI it is excluded from the DOI count even if its arXiv id resolves to one upstream. The temporal coverage of those papers is summarised by fig. 2, and their backend-reported relevance scores by fig. 3.

## 4.2 Output artefacts

After running `scripts/run_search_pipeline.py` against the default `manuscript/config.yaml`, the project produces:

- `output/search/results.json` — the raw `SearchResult` JSON, including `per_source_counts` and `errors` for diagnostic purposes.
- `output/search/cache/search_<hash>.json` — the deterministic search cache; identical reruns are file reads.
- `output/cache/abs/<safe_id>.txt` — one file per fetched abstract.
- `output/cache/pdf/<safe_id>.{pdf,txt}` — PDFs and extracted text (only when `enrichment.fetch_fulltext: true`).
- `output/corpus.json` — a `LocalBackend`-compatible JSON corpus of every result, enriched in place.
- `manuscript/references.bib` — the auto-populated bibliography from the single-query pipeline (merged with any other `manuscript/*.bib` at PDF render time).
- `output/llm/per_paper/<safe_id>.md` — per-paper LLM analyses (only when `llm.per_paper: true` *and* the LLM stack is reachable).
- `output/llm/synthesis.md` — corpus-level LLM synthesis (only when `llm.corpus_synthesis: true` *and* the LLM stack is reachable).

- `output/reading_report.md` — the final assembled reading report.

When the LLM stack is genuinely unreachable, the `output/llm/` artefacts are simply absent — no placeholder file is ever written into the archive (see sec. 6).

Because the search cache and abstract cache are deterministic, a second run with identical `config.yaml` produces byte-identical artifacts (modulo timestamp metadata in the cache files themselves). This is the property the project exists to demonstrate.

The exact paper count, DOI list, and synthesis text depend on the live state of arXiv and Crossref at the time of the run — and are therefore not reproducible *across* runs in different weeks. Users seeking strict reproducibility should:

1. Pin a `LocalBackend` corpus generated from a successful run (`infrastructure.search.literature.write_corpus`) and remove `arxiv` / `crossref` from `project_config.search.sources`.
2. Commit the `output/search/cache/` directory to version control.
3. Pin the LLM seed (`config.llm.seed`) and avoid model upgrades.

With those three steps, every run from the same commit produces the same outputs.

# 5 Conclusion

`template_search_project` packages a complete, configurable, reproducible literature workflow into the Research Project Template's two-layer architecture. By keeping discovery, export, and synthesis in three orthogonal infrastructure modules, the project demonstrates that ambitious research automation can still respect the template's principles:

- **Single source of truth** — `Paper` for discovery, `BibEntry` for export, structured `SynthesisResult` records for LLM output.
- **Test-driven development** — every module is covered by real-data tests; HTTP backends are exercised through `pytest-httpserver`, the LLM bridge through deterministic local callables.
- **Thin orchestrator pattern** — `scripts/run_search_pipeline.py` does only argument parsing, configuration loading, and I/O; all logic lives in `infrastructure/` or `src/`.
- **No mocks** — neither in the new infrastructure modules nor in the project test suite.
- **Multi-project support** — the project lives alongside `template_code_project/` and follows the same layout, so the existing pipeline runner discovers and executes it without modification.
- **Reproducibility** — deterministic search caching, on-disk enrichment caching, and pinned LLM seeds make a single `manuscript/config.yaml` the only artifact a reviewer needs.

We close with three concrete extensions that build naturally on this foundation:

1. **Crossref TDM full-text fetch** for non-arXiv DOIs, completing the abstract-to-fulltext picture without changing the project's API.
2. **CSL-JSON export** alongside BibTeX, enabling Zotero / Mendeley / Pandoc-CSL workflows from the same `BibDatabase`.
3. **Vector recall on `LocalBackend`** for sufficiently large curated corpora, gated behind an optional dependency and an explicit benchmark-derived crossover threshold.

The infrastructure modules are deliberately small and stable; the project that exercises them is deliberately small and explicit. Together they show that *domain-specific research automation* and *template-strict architectural discipline* are compatible — and, in fact, mutually reinforcing.

The bundled `data/corpus.json` is a deterministic workflow fixture containing citation-shaped optimisation records [Boyd and Vandenberghe, 2004, Nocedal and Wright, 2006, Nesterov, 2013, Kingma and Ba, 2014, Reddi et al., 2018, Peng, 2011]. It ensures that the auto-generated `manuscript/references.bib` contains citation-ready entries that downstream tooling can resolve; it does not establish empirical literature findings.

# 6 Pipeline Internals

This supplemental section documents the data structures and on-disk artifacts the pipeline produces, for readers who want to extend or audit it.

## 6.1 Data structures

The Mermaid class diagram in this subsection shows the canonical fields each record carries through the pipeline. Records have additional optional metadata (e.g. `Paper.url`, `Paper.publisher`, `Paper.isbn`, `Paper.raw`) omitted for readability — consult `infrastructure/search/literature/models.py` (source on GitHub) and `src/pipeline/pipeline.py` (source on GitHub) for the full schema.

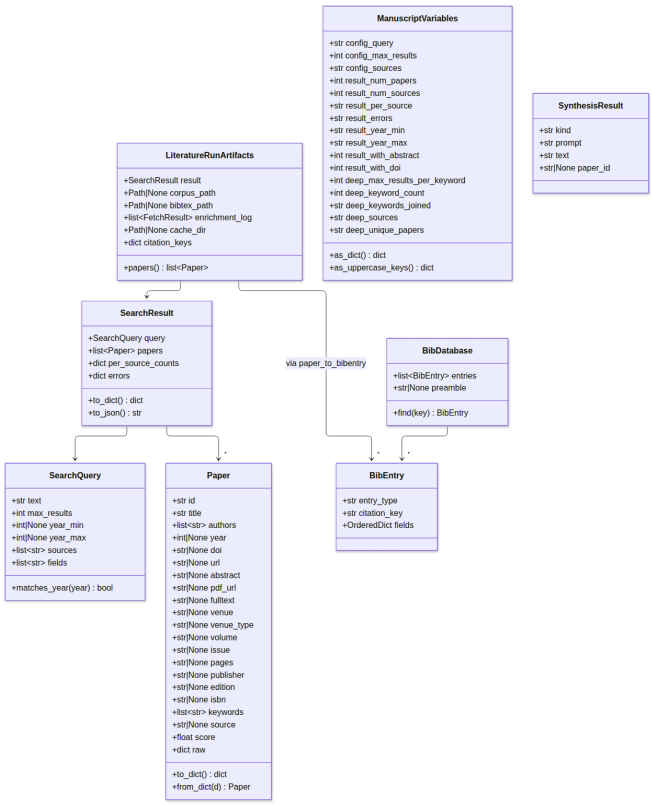



Figure 4: Mermaid diagram

## 6.2 On-disk layout

The project keeps committed input data in `data/`, regeneratable outputs in `output/` (gitignored), and the manuscript source in `manuscript/`. The Mermaid flowchart in this subsection illustrates every artefact the standard pipeline writes:

- `data/corpus.json` — the bundled offline corpus, CI-safe default for `LocalBackend`.
- `output/search/results.json` — raw `SearchResult` JSON from the latest run.
- `output/search/cache/search_<hash>.json` — deterministic `SearchCache` files.
- `output/cache/abs/<safe_id>.txt` — one cached abstract per paper.
- `output/cache/pdf/<safe_id>.{pdf,txt}` — PDF bytes plus extracted text.
- `output/llm/synthesis.md` — corpus-level LLM synthesis (when enabled).
- `output/llm/per_paper/<safe_id>.md` — one per-paper note per paper (when enabled).
- `../figures/{papers_per_source,year_histogram,score_distribution}.png` — diagnostic figures.

- `output/data/manuscript_variables.json` — substitution table consumed by the resolver.
- `output/corpus.json` — enriched corpus, written in `LocalBackend`-compatible format so it can re-seed a deterministic future run.
- `output/enrichment_log.json` — one entry per fetcher per paper.
- `output/reading_report.md` — final markdown reading report.
- `output/run_summary.json` — one-line metadata for the run.
- `manuscript/references.bib` — auto-populated, Pandoc-ready BibTeX.

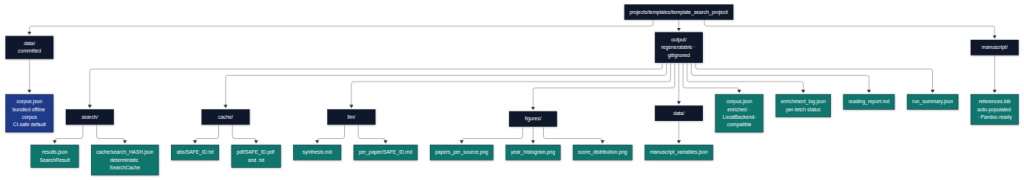


Figure 5: Mermaid diagram

## 6.3 Citation-key collision handling

`paper_to_bibentry()` generates citation keys as `<author><year><title-word>` (with stop-words filtered and unicode folded). When two papers in the same result set produce the same key — common when one author publishes multiple papers in the same year on closely related topics — `src/pipeline/pipeline.py::_disambiguate_citation_key` appends a deterministic suffix from the alphabet (`a`, `b`, …, `z`, then two-letter combinations `aa`, `ab`, …) until uniqueness is restored, with a numeric `_1`, `_2`, … fallback for the pathological case. The mapping is exposed to downstream stages via `LiteratureRunArtifacts.citation_keys`, and the report uses these keys verbatim, so the LLM synthesis and the BibTeX file always agree.

The deep-search workflow has its own collision handler in `src/search/deep_search.py::run_deep_search` that operates over the post-deduplication aggregate roster — see sec. 8 — and the unified `references_deep.bib` reflects the same mapping.

## 6.4 Failure isolation

- **A backend is unreachable.** `LiteratureClient` records the message into `result.errors[name]` and continues. The reading report surfaces these errors in a callout block at the top.
- **An abstract fetch fails.** The fetcher records `status="error"` in `enrichment_log.json` and the paper keeps its existing (possibly empty) abstract.
- **A PDF fetch fails or `pypdf` is missing.** Same pattern — the PDF, if downloaded, is still cached on disk. The reading report does not reference the missing fulltext.
- **The LLM is unreachable or unconfigured.** `scripts/run_search_pipeline.py` logs a warning and skips the synthesis stage entirely — `output/llm/` is left empty and the reading report omits both the per-paper-notes and cross-corpus sections. No placeholder text is ever written into the archive, so a missing LLM is observable from the absence of those sections rather than from a fake "(LLM unavailable)" string.

# 7 Reproducibility

Reproducibility in computational research has well-documented prerequisites: open data, open code, and a deterministic build that can be re-run from scratch [Peng, 2011]. The bundled `manuscript/config.yaml` is intentionally configured to satisfy all three for **strict reproducibility**:

1. `project_config.search.sources: [local]` consumes `data/corpus.json`, which is a curated and committed JSON corpus. No network is required to run the pipeline.
2. `search.cache_dir: output/search/cache` writes deterministic JSON cache files; running the same query twice produces a byte-identical artifact tree (modulo timestamp metadata in the cache file itself).
3. `enrichment.fetch_abstracts: true` reads abstracts directly from the corpus when present; no network fetch is required.
4. `enrichment.fetch_fulltext: false` is the default — full-text fetching is opt-in and gated behind the optional `pypdf` dependency (`uv sync --group rendering`).
5. `llm.enabled: false` is the default — the LLM stage is opt-in and requires a running `ollama serve`. When enabled, `seed: 42` and `temperature: 0.0` are pinned.

## 7.1 Switching to live search

Replace the `sources` list with the desired backend set:

```yaml
project_config:
  search:
    query: "your topic"
    sources: [arxiv, crossref]
    crossref_mailto: "you@example.org"
```

A first live run populates `output/search/cache/`. Commit that directory to the repo and the pipeline becomes reproducible across machines without further configuration changes.

## 7.2 Determinism guarantees

The full determinism contract is itemised in tbl. 1: every pipeline stage is annotated as fully, conditionally, or non-deterministic, with an explicit mechanism column so reviewers can audit each row independently.

Table 1: Determinism contract by pipeline stage. Cached stages are byte-stable across reruns; live stages depend on the upstream source and are pinned to the cache file once a successful run completes.

| Stage | Deterministic? | Mechanism |
|---|---|---|
| Search (cached hit) | yes | `SearchCache` JSON files |
| Search (cache miss) | no | live API |
| Dedup / merge | yes | DOI / arXiv-id canonical keys; tie-break by score then year |
| Citation-key generation | yes | unicode folding + stop-word skip; collision suffix is deterministic |
| BibTeX writer | yes | byte-stable format pinning (verified by `tests/infra_tests/reference/`) |
| Abstract fetch | yes (cached) / no (live) | per-paper `<safe_id>.txt` cache |
| Fulltext fetch | yes (cached) / mostly (live) | per-paper `<safe_id>.{pdf,txt}` cache; live fetch's `pypdf` text extraction is not bit-stable across versions |
| LLM synthesis | mostly | `seed=42`, `temperature=0.0`; Ollama deterministic up to its own minor variance |

| Stage | Deterministic? | Mechanism |
|---|---|---|
| Figure generation | yes (within Matplotlib version) | fixed palette, fixed bin width, no random subsampling |

## 7.3 Verifying reproducibility locally

```
# Run twice; nothing in output/ should diff except the cache timestamps.
uv run python projects/templates/template_search_project/scripts/run_search_pipeline.py
mv projects/templates/template_search_project/output
projects/templates/template_search_project/output_first
uv run python projects/templates/template_search_project/scripts/run_search_pipeline.py
diff -ru \
    projects/templates/template_search_project/output_first/corpus.json \
    projects/templates/template_search_project/output/corpus.json
```

The only expected differences are inside `output/search/cache/search_*.json`, where `_cached_at` is wall-clock time at write.

## 7.4 Limitations

The reproducibility contract enumerated in sec. 7 (items 1–5 and tbl. 1) does not eliminate the following well-defined sources of non-reproducibility, which are surfaced here so reviewers can audit them explicitly rather than inferring from the contract table:

- **Live search drift.** When `project_config.search.sources` includes `arxiv` or `crossref`, the first cache-miss invocation hits the live API; the cached JSON freezes that response, but two cold-start clones running on different days will see different paper sets. Pin a `LocalBackend` corpus or commit `output/search/cache/` to break this dependency.
- **`pypdf` version drift.** The fulltext fetcher uses `pypdf` to extract text from a downloaded PDF. `pypdf`'s text-extraction algorithm is not bit-stable across major versions; upgrading `pypdf` can produce different `<safe_id>.txt` cache contents from the same source PDF. The PDF bytes themselves are bit-stable so the cache freezes the inputs, not the extraction.
- **Ollama version drift.** Pinning `seed=42` and `temperature=0.0` controls Ollama's sampling, but the model weights, tokenizer, and template can change between Ollama releases. Document the Ollama version alongside `config.llm.model` when archiving a run for replication.
- **Paperclip backend status.** The `paperclip` backend is opt-in and the production endpoint may reject the adapter's request method; the run records the response in `SearchResult.errors[paperclip]` and continues. Treat `paperclip` results as advisory until the upstream service stabilises.
- **External backend behaviour outside this project's control.** arXiv and Crossref are the source of truth; this project is faithful to whatever they return. A retraction, metadata fix, or DOI assignment upstream will alter the cache on the next cold-start invocation.

These limitations bound *what* the cache + seed + corpus pinning achieves. Inside those bounds, the contract in tbl. 1 is total: every cached pipeline stage is byte-stable across reruns (verified by `tests/test_pipeline.py::TestRunLiteraturePipeline::test_bibtex_byte_identical_across_reruns`).

# 8 Deep Search

The deep-search workflow extends the standard literature pipeline along three axes: **breadth** (multi-keyword fan-out), **depth** (full enrichment of every paper, abstract + PDF fulltext), and **archival quality** (per-paper multi-section LLM reading notes saved as standalone markdown). It is invoked from the same configuration file but a separate orchestrator script.

## 8.1 Configuration

The `deep_search:` block in `manuscript/config.yaml` controls the run. The most-used knobs:

- `keywords` — list of free-text queries (each becomes one `SearchQuery`).
- `max_results_per_keyword` — per-keyword cap (100 by default; honoured by the aggregator after dedup).
- `sources` — backend list, same vocabulary as the standard pipeline (`arxiv`, `crossref`, `local`, `paperclip`).
- `fetch_abstracts` / `fetch_fulltext` — when both are `true`, every returned paper has its abstract fetched (arXiv export API) and (where a PDF URL is available) its fulltext extracted via `pypdf`. Cached on disk under `output/cache/abs/` and `output/cache/pdf/`.
- `llm_per_paper` — when `true` and Ollama is reachable, each paper gets a multi-section markdown reading note generated by the local LLM. When `false`, or when the LLM stack is genuinely unreachable at runtime, the per-paper note is written with only the abstract / fulltext-excerpt sections; the synthesis section is omitted entirely (no placeholder text). The composer's Supplemental S1 also drops the per-paper synthesis subsection in that case rather than emitting empty rows.
- `write_unified_bibtex` / `unified_bibtex_path` — when `true`, a deduplicated, collision-suffixed BibTeX file is written under `manuscript/references_deep.bib` so it can be cited from the manuscript via Pandoc `[@key]` syntax exactly the same way as the hand-curated `references.bib`.

## 8.2 Pipeline shape

The deep-search pipeline reads `project_config.deep_search` from `manuscript/config.yaml`, runs one `SearchQuery` per keyword (capped at `max_results_per_keyword`), aggregates the per-backend results through `LiteratureClient`, enriches every paper via `AbstractFetcher` and `FulltextFetcher`, generates collision-free citation keys with `paper_to_bibentry`, and (when `llm_per_paper: true` and Ollama is reachable) calls the local LLM with `DEEP_PROMPT` to produce a seven-section reading note per paper. The per-keyword outputs are then merged by `merge_papers` to form a deduplicated aggregate roster which is written to `manuscript/references_deep.bib`, `output/deep_search/aggregate.json`, and `output/deep_search/aggregate_report.md`. A Mermaid flowchart in this subsection illustrates the same flow.

## 8.3 LLM prompt

The deep-search prompt is richer than the standard `synthesis.PROMPT_PER_PAPER`. It produces a self-contained reading note suitable for archival without re-reading the paper:

```
## Contribution
One paragraph stating the paper's central claim and why it is novel.

## Method
2-4 bullets describing the technical approach.

## Evidence
2-4 bullets describing experiments / proofs.

## Limitations
1-3 bullets covering caveats and what the paper does NOT address.
```

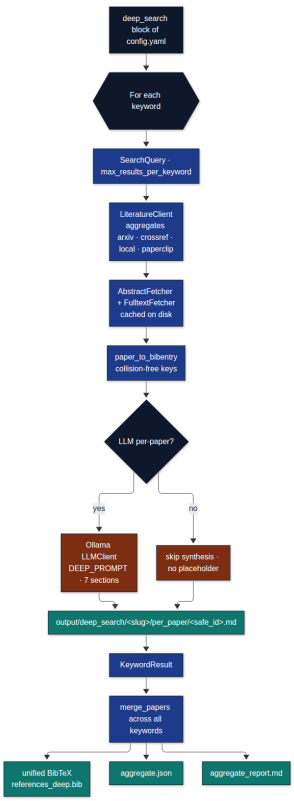



Figure 6: Mermaid diagram

```
## Connections
1-3 bullets relating this paper to other work in the field, citing only
papers explicitly named in the input.

## Significance for {keyword}
One paragraph explaining why this paper matters for the keyword.

## Tags
5-10 lowercase keywords.
```

The LLM is duck-typed as `Callable[[str], str]` so tests pass a deterministic local callable that returns real, well-formed reading-note text; runtime callers wrap an `infrastructure.llm.LLMClient` with `seed=42, temperature=0.0` for reproducibility. When the LLM stack is unreachable, callers pass `None` and the per-paper synthesis stage is skipped entirely — no placeholder text is ever written into the archive.

## 8.4 On-disk layout

A successful deep-search run writes the following artefacts (paths shown relative to the project root); a Mermaid tree in this subsection depicts the same hierarchy:

- `output/deep_search/aggregate.json` — every unique paper across keywords.
- `output/deep_search/aggregate_report.md` — cross-keyword markdown summary.
- `output/deep_search/run_summary.json` — one-line metadata for the run.
- `output/deep_search/<keyword_slug>/papers.json` — enriched per-keyword paper list.
- `output/deep_search/<keyword_slug>/reading_report.md` — per-keyword markdown summary.
- `output/deep_search/<keyword_slug>/per_paper/<safe_id>.md` — one reading note per paper.
- `manuscript/references_deep.bib` — auto-generated, deduplicated unified BibTeX file.

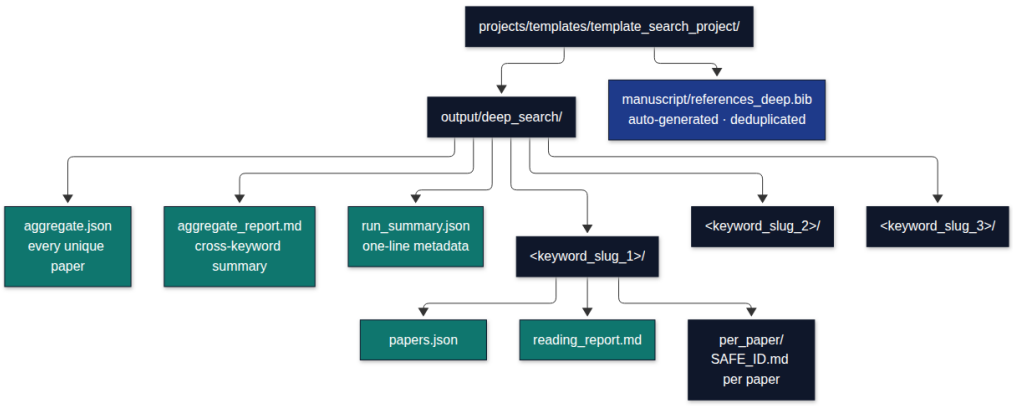



Figure 7: Mermaid diagram

## 8.5 Determinism

Three caches make this workflow replayable:

1. `SearchCache` (`output/search/cache/search_<hash>.json`) — keyed on canonical query identity per keyword.
2. Abstract cache (`output/cache/abs/<safe_id>.txt`).
3. Fulltext cache (`output/cache/pdf/<safe_id>.{pdf,txt}`).
4. LLM seed (`project_config.deep_search.llm_seed: 42`) + temperature (`0.0`).

Commit any subset of these caches to version control to freeze a run.

## 8.6 Paperclip backend

The Paperclip backend is opt-in. Enable it by:

1. Creating `projects/templates/template_search_project/.env` (gitignored) with your `PAPERCLIP_API_KEY=gxl_…` (template at `.env.example`).
2. Including `paperclip` in `project_config.deep_search.sources`.

The orchestrator scripts auto-load `.env` via the lightweight `src.dotenv` module before constructing the backend list. The `PaperclipBackend` adapter mirrors the wire protocol of the official `gxl_paperclip` Python SDK: `POST /papers` with an `X-API-Key` header and a JSON-RPC `tools/call` envelope. We support both the modern `structuredContent.papers` response shape and the older text-only `content[].text` shape (best-effort regex extraction).

Failure-isolation: any per-backend error (including a method-rejection response the production endpoint may return during migration) is captured into `SearchResult.errors[paperclip]` and the run continues. See `output/deep_search/<keyword>/papers.json` and the aggregate `run_summary.json` for the recorded errors.

## 8.7 CLI

```
# Run with everything from config.yaml (assumes project_config.deep_search.enabled: true)
uv run python projects/templates/template_search_project/scripts/run_deep_search.py

# Force-enable without touching config.yaml
uv run python projects/templates/template_search_project/scripts/run_deep_search.py
--enable

# Override keyword list at the command line
uv run python projects/templates/template_search_project/scripts/run_deep_search.py \
    --enable --keyword "convex optimization" --keyword "stochastic gradient descent"

# Skip LLM stage even when config enables it
uv run python projects/templates/template_search_project/scripts/run_deep_search.py
--enable --no-llm

# Bypass cache (writes still happen)
uv run python projects/templates/template_search_project/scripts/run_deep_search.py
--enable --no-cache

# Local-corpus mode (offline · CI-friendly)
uv run python projects/templates/template_search_project/scripts/run_deep_search.py \
    --enable --corpus projects/templates/template_search_project/data/corpus.json
```

# 9 Supplemental S1 — Literature Review (auto-composed from deep search)

*Composed by `scripts/s_compose_literature_review.py` from the most recent deep-search run. Edit the script, not this file — manual edits will be overwritten on the next pipeline run.*

This review covers **3 keyword(s)**, **300 unique paper(s)** (retrieved at up to 100 per keyword from arxiv, crossref). All references are stored in `manuscript/references_deep.bib` and resolved by the combined-PDF pipeline (Pandoc `--natbib` + BibTeX over all `manuscript/*.bib` files).

## 9.1 convex optimization

*Papers retrieved:* **100** · *per-source contributions:* arxiv=100, crossref=100 · *backend errors:* none

| Cite | Title | Year | DOI / URL |
|---|---|---|---|
| [Cambridge University Press, 2004b] | Convex optimization problems | 2004 | DOI 10.1017/cbo9780511804441.005 (open) |
| [Cambridge University Press, 2004c] | Convex functions | 2004 | DOI 10.1017/cbo9780511804441.004 (open) |
| [Springer-Verlag, e] | Reverse Convex Optimization, Reverse convex programming | n.d. | DOI 10.1007/springerreference_72642 (open) |
| [Suh, 2022] | 1. Convex Optimization Basics | 2022 | DOI 10.1561/9781638280538.ch1 (open) |
| [Cambridge University Press, 2021a] | Convex Optimization and Efficiency | 2021 | DOI 10.1017/9781108699211.006 (open) |
| [Cambridge University Press, 2021d] | Ellipsoid Method for Convex Optimization | 2021 | DOI 10.1017/9781108699211.015 (open) |
| [Nesterov, 2004b] | Smooth Convex Optimization | 2004 | DOI 10.1007/978-1-4419-8853-9_2 (open) |
| [Nesterov, 2004a] | Nonsmooth Convex Optimization | 2004 | DOI 10.1007/978-1-4419-8853-9_3 (open) |
| [Emerald Publishing Limited, 2022] | Convex Optimization Basics | 2022 | DOI 10.1561/978-1-63828-053-820251002 (open) |
| [Cambridge University Press, 2004d] | Convex sets | 2004 | DOI 10.1017/cbo9780511804441.003 (open) |
| [CRC Press, 2011g] | What Is Convex Optimization? | 2011 | DOI 10.1201/b11156-7 (open) |
| [CRC Press, 2011c] | Convex Semi-Infinite Optimization | 2011 | DOI 10.1201/b11156-17 (open) |
| [CRC Press, 2011d] | Tools for Convex Optimization | 2011 | DOI 10.1201/b11156-8 (open) |
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| [Princeton University Press, 2020a] | Appendix: | 2020 | DOI 10.2307/j.ctvqsdxqd.15 (open) |
| [*, 2005] | Connectedness of efficient points in convex and convex transformable vector optimization | 2005 | DOI 10.1080/02331930500096270 (open) |
| [Cambridge University Press, 2025c] | Convex Programming Problems and Convex Theorem of the Alternative | 2025 | DOI 10.1017/9781009510561.021 (open) |
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## 9.2 stochastic gradient descent

*Papers retrieved:* **100** · *per-source contributions:* arxiv=100, crossref=100 · *backend errors:* none

| Cite | Title | Year | DOI / URL |
|---|---|---|---|
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## 9.3 reproducible research

*Papers retrieved:* **100** · *per-source contributions:* arxiv=100, crossref=100 · *backend errors:* none

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*Per-paper synthesis omitted — no LLM Contribution paragraphs are present in the deep-search outputs (set `project_config.deep_search.llm_per_paper: true` and ensure Ollama is reachable to populate this section).*

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*Composition summary: 3 keywords · 300 unique papers · 300 per-paper notes integrated · 300 BibTeX entries · 0 key(s) missing from bib.*

# 10 References

This project can produce **two** BibTeX files; the template combined-PDF path uses Pandoc `--natbib` plus BibTeX and merges every `manuscript/*.bib` for citation resolution:

- `manuscript/references.bib` — single-query pipeline output (`scripts/run_search_pipeline.py`).
- `manuscript/references_deep.bib` — deduplicated multi-keyword deep-search output (`scripts/run_deep_search.py`). Every citation in sec. 9 resolves against this file. The supplemental section is auto-composed by `scripts/s_compose_literature_review.py`; do not edit by hand.

To regenerate the standard bibliography:

```
uv run python projects/templates/template_search_project/scripts/run_search_pipeline.py
```

To regenerate the deep-search bibliography (10 papers per keyword, fully enriched, LLM-summarised):

```
uv run python projects/templates/template_search_project/scripts/run_deep_search.py
uv run python
projects/templates/template_search_project/scripts/s_compose_literature_review.py
```

To validate that either `.bib` is syntactically clean and contains the required fields per entry type:

```
uv run python -m infrastructure.reference.citation.cli validate \
    projects/templates/template_search_project/manuscript/references.bib --strict
uv run python -m infrastructure.reference.citation.cli validate \
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